

Polynomials – 10

Where the roots lie: bounds, Gauss–Lucas, stability and
real-rootedness

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Theoretical minimum. The position of the roots can be read off without solving the equation, by two devices: comparing the sizes of the monomials (a dominant term forbids roots or forces a sign), and the logarithmic derivative $P'/P = \sum_j \frac{1}{z-z_j}$, whose real or imaginary part has a fixed sign away from the roots. Counting real roots (Descartes, Sturm, Newton) is done in Polynomials – 4, the argument principle and Rouché in Complex Analysis – 3.

Definitions. $P(z) = a_n z^n + \dots + a_1 z + a_0$ with $a_n \neq 0$, and $D = \frac{d}{dx}$. A real polynomial is *real-rooted* if all its roots are real, and (*Hurwitz*) *stable* if all its roots satisfy $\operatorname{Re} z < 0$. Two real-rooted polynomials f, g with $\deg f - \deg g \in \{0, 1\}$ and simple roots *interlace* if their roots alternate on the line: between two consecutive roots of one of them there is exactly one root of the other.

Theorem 1 (bounds from the coefficients). a) (Cauchy) Unless $a_0 = \dots = a_{n-1} = 0$, every root satisfies $|z| \leq R$, where R is the unique positive root of $|a_n|x^n = |a_{n-1}|x^{n-1} + \dots + |a_0|$; hence $|z| < 1 + \max_{k < n} |a_k/a_n|$ and (Fujiwara) $|z| \leq 2 \max_{1 \leq k \leq n} |a_{n-k}/a_n|^{1/k}$. b) (Dominant monomial) If $|a_m|r^m > \sum_{k \neq m} |a_k|r^k$, then P has no roots on $|z| = r$ and exactly m roots in $|z| < r$ (Rouché). For real x the same inequality with $r = |x|$ gives $\operatorname{sign} P(x) = \operatorname{sign}(a_m x^m)$: points at which different monomials dominate produce sign changes, hence real roots, by the intermediate value theorem alone. c) (Eneström–Kakeya) If all $a_k > 0$, every root satisfies $\min_k \frac{a_k}{a_{k+1}} \leq |z| \leq \max_k \frac{a_k}{a_{k+1}}$; in particular $0 < a_0 \leq a_1 \leq \dots \leq a_n$ gives $|z| \leq 1$. *Proof sketch.* a) For $|z| > R$ the leading term beats all the others together, and $x \mapsto \sum_{k < n} |a_k|x^{k-n}$ is decreasing; for Fujiwara check the inequality at $x = 2M$. c) In the monotone case $(1-z)P(z) + a_n z^{n+1} = a_0 + \sum_{k \geq 1} (a_k - a_{k-1})z^k$ has modulus at most $a_n|z|^n$ for $|z| \geq 1$; in general put $z = \rho w$ with $\rho = \max a_k/a_{k+1}$, and use $z^n P(1/z)$ for the lower bound.

Theorem 2 (Gauss–Lucas). The roots of P' lie in the convex hull of the

roots $z_1, \dots, z_n$ of P . *Proof.* If $P'(w) = 0 \neq P(w)$, then $0 = \overline{P'(w)/P(w)} = \sum_j \frac{w - z_j}{|w - z_j|^2}$, so w is a convex combination of the z_j with weights proportional to $|w - z_j|^{-2}$. So every disc, half-plane or strip containing the roots of P contains those of $P', P'', \dots$ (Complex Analysis – 1). The same computation gives more: if all $\operatorname{Re} z_j \leq c$, then $\operatorname{Re}(P'/P) > 0$ on $\{\operatorname{Re} z > c\}$; if all $\operatorname{Im} z_j \leq c$, then $\operatorname{Im}(P'/P) < 0$ on $\{\operatorname{Im} z > c\}$.

Theorem 3 (stable polynomials). a) Every real stable polynomial is a product of factors $z + \alpha$ and $z^2 + pz + q$ with $\alpha, p, q > 0$, so all its coefficients are non-zero and of one sign. Products of stable polynomials are stable, and if f is stable, then so are f' and $f + af'$ for $a \geq 0$, since $\operatorname{Re}(1 + af'/f) > 0$ on $\{\operatorname{Re} z \geq 0\}$. b) (Routh–Hurwitz in small degree) Let all $a_k > 0$. For $n \leq 2$ the polynomial is stable; for $n = 3$ it is stable iff $a_1 a_2 > a_0 a_3$; for $n = 4$ iff $a_1 a_2 a_3 > a_0 a_3^2 + a_4 a_1^2$. In general: iff all leading principal minors of the Hurwitz matrix $(a_{n-2j+i})_{i,j=1}^n$ are positive (may be quoted). *Proof idea.* The roots depend continuously on the coefficients and can leave the left half-plane only through $i\mathbb{R}$; the equation $P(iy) = 0$ splits into the even and the odd part, and eliminating y gives exactly the boundary $a_1 a_2 = a_0 a_3$ (resp. the quartic one). Test one polynomial on each side, e.g. $(z + 1)^n$.

Theorem 4 (operators preserving real-rootedness). Let f be real-rooted. a) For every real a the polynomial $f + af'$ is real-rooted: by Theorem 2, $\operatorname{Im}(f'/f) \neq 0$ off $\mathbb{R}$. Likewise, a strip $\{|\operatorname{Im} z| \leq c\}$ containing the roots of f contains those of $f + af'$. (Another proof: $f + af' = ae^{-x/a}(e^{x/a}f)'$ and Rolle, Polynomials – 4.) b) (Hermite–Poulain) If $\varphi(t) = \sum_k b_k t^k$ is real-rooted, then $\varphi(D)f = \sum_k b_k f^{(k)}$ is real-rooted or 0: factor φ into t 's and $(1 + at)$'s. c) (Laguerre) If $f = \sum_{k=0}^n c_k x^k$ and φ is real-rooted with no roots in the open interval $(0, n)$, then $\varphi(xD)f = \sum_k \varphi(k) c_k x^k$ is real-rooted or 0: use $xf' - \gamma f = x^{\gamma+1}(x^{-\gamma}f)'$ on each half-line and Rolle. d) (Hermite–Kakeya–Obreschkoff) For real f, g without common roots and with $\deg f - \deg g \in \{0, 1\}$, all combinations $\lambda f + \mu g$, $(\lambda, \mu) \in \mathbb{R}^2 \setminus \{0\}$, are real-rooted iff f and g have simple real interlacing roots. *Idea:* both conditions say that g/f takes every value in $\mathbb{R} \cup \{\infty\}$ exactly $\deg f$ times on $\mathbb{R} \cup \{\infty\}$, i.e. is monotone between its poles. Example: f and f' for f with simple real roots.

Fact 1 (Descartes for exponential sums). For real $\lambda_1 < \dots < \lambda_m$ and non-zero real c_i , the function $F(x) = \sum_i c_i e^{\lambda_i x}$ has at most as many

real zeros, counted with multiplicity, as there are sign changes in $c_1, \dots, c_m$. *Proof.* If the sign changes between c_j and c_{j+1} , take $\lambda_j < \lambda < \lambda_{j+1}$: then $(e^{-\lambda x} F)'$ has one sign change fewer, and apply Rolle. For $\lambda_i \in \mathbb{Z}_{\geq 0}$ and $x = \ln t$ this is Descartes' rule (Polynomials – 4).

1. Prove Theorem 1 in full. Deduce that for every $n \geq 1$ all roots of $1 + z + \frac{z^2}{2!} + \dots + \frac{z^n}{n!}$ lie in the annulus $1 \leq |z| \leq n$, and show that the Cauchy radius R cannot be improved: for any given moduli $|a_0|, \dots, |a_n|$ some polynomial with coefficients of these moduli has a root of modulus exactly R .

2. Let n be a positive integer. Show that there are positive real numbers $a_0, a_1, \dots, a_n$ such that for each choice of signs the polynomial

$$\pm a_n x^n \pm a_{n-1} x^{n-1} \pm \dots \pm a_1 x \pm a_0$$

has n distinct real roots. (IMC 2014)

3. Let p and q be complex polynomials with $\deg p > \deg q$ and let $f(z) = \frac{p(z)}{q(z)}$. Suppose that all roots of p lie inside the unit circle $|z| = 1$ and that all roots of q lie outside the unit circle. Prove that

$$\max_{|z|=1} |f'(z)| > \frac{\deg p - \deg q}{2} \max_{|z|=1} |f(z)|.$$

(VJIMC 2011, cat. II)

4. Let $f(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$ be a polynomial with real coefficients. Prove that if all roots of f lie in the left half-plane $\{z \in \mathbb{C} : \operatorname{Re} z < 0\}$, then

$$a_k a_{k+3} < a_{k+1} a_{k+2}$$

holds for every $k = 0, 1, \dots, n-3$. (IMC 2003)

5. Let $f \neq 0$ be a polynomial with real coefficients. Define the sequence $f_0, f_1, f_2, \dots$ of polynomials by $f_0 = f$ and $f_{n+1} = f_n + f'_n$ for every $n \geq 0$. Prove that there exists a number N such that for every $n \geq N$ all roots of f_n are real. (IMC 2007)